Ultra-Wideband Dynamic Swarm Ranging

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Abstract

Ultra-wideband (UWB) ranging technology is increasingly being applied to measure distances among dynamic embodied agents. However, the classic DS-TWR protocol was originally designed for static ranging and therefore suffers from ranging latency problem when deployed in high-speed dynamic multi-agent swarms. We establish a novel *iterative convergent theory* for such scenario and propose to compensate through prediction, and thus design the Dynamic Swarm Ranging (DSR) protocol that directly addresses the ranging latency problem. DSR is significantly different from DS-TWR, representing a fundamental theoretical breakthrough in a decade-old, well-established, and widely adopted ranging theory. We design a dedicated ranging message, ranging table (data structure) and a novel state machine and its transition that maintains ranging table updates. Moreover, our implemented ranging message format maintains compatibility with TDoA, enabling DSR and TDoA to seamlessly switch or coexist. Extensive experiments demonstrate that the proposed method achieves significant improvements over DS-TWR. In dynamic scenarios, DSR reduces Mean Absolute Error (Mean AE) by 37.62% and Root Mean Square Error (RMSE) by 32.10%, confirming its effectiveness for accurate and real-time ranging in multi-agent dynamic environments.

Keywords

Ultra-wideband, Dynamic swarm ranging, Double-sided two-way ranging, Time-of-Flight, Protocol Design

1 Introduction

Ultra-wideband (UWB) technology, originating from impulse radio research in the 1960s, gained significant attention after the Federal Communications Commission (FCC) allocated unlicensed spectrum for UWB in 2002. By leveraging nanosecond-scale impulse transmissions, UWB enables highly accurate timestamping of signal transmission and reception, thereby allowing precise distance estimation based on Time-of-Flight (ToF) measurements. Owing to its high temporal resolution and robustness to multipath effects, UWB has been widely applied in ranging [20, 27, 36, 37], localization [15, 26, 38, 47, 49, 51], robotics [23, 35, 46, 50], logistics tracking [7, 10, 12, 21], human tracking [6, 11, 17], sensing [25, 41, 42], and orientation estimation [52].



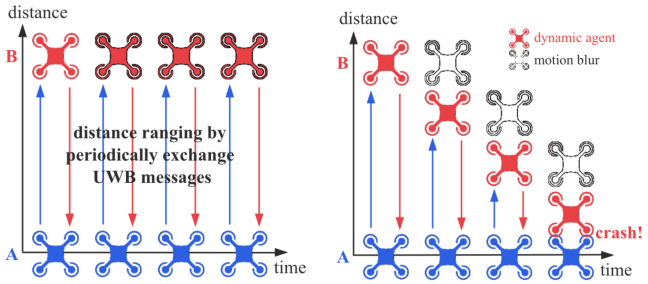
Figure 1: A new application trend of UWB is distance ranging among a group of dynamic embodied intelligent agents, which is driven by the rapid deployment of humanoid, wheeled, legged, and aerial robots. These agents typically operate under high dynamics and collaborate to accomplish complex tasks in groups. Accurate and reliable distance ranging becomes critical, as it serves as the foundation for localization, formation maintenance, and collision avoidance.

Common UWB ranging techniques include Two-Way Ranging (TWR), Time Difference of Arrival (TDoA), and Phase-Difference-of-Arrival (PDoA). The mainstream UWB ranging technique is the **double-sided two-way ranging (DS-TWR)**, which was proposed in 2016 by researchers from industry [30]. Over the past decade, DS-TWR has been standardized [1, 2], commercialized [5, 9, 16, 31, 40], and extensively studied in academia [20, 23, 27, 35–37, 50].

Four years after its proposal, in 2020, the DS-TWR was officially standardized in the IEEE 802.15.4z standard [1] and was subsequently included in the white paper released by FiRa in 2022 [2], a consortium of industrial companies. Since then, UWB devices compliant with IEEE 802.15.4z or FiRa specifications have been widely commercialized, including the iPhone and Apple Watch [5], the BMW [9] and Cadillac [16] vehicles, and the Unitree Go1 and Go2 robots [40].

In the academic domain, DS-TWR has seen several extensions and variations. In 2021, it was applied to dynamic and dense networks [36, 37]. In 2022, a variant called V-TWR was proposed to enable efficient overhearing for large-scale tag deployments [47]. In 2024, another variant was developed for multi-robot relative pose estimation [35], followed in 2025 by a version targeting ranging-based relative localization [23]. Most recently, in 2026, the performance limits of DS-TWR in dynamic and dense networks were analyzed [20].

A clear application trend of UWB is distance ranging among a group of dynamic embodied intelligent agents [20, 23, 35–37, 47, 49, 51], as in Fig. 1. This trend is driven by the rapid deployment of humanoid, wheeled, legged, and



(a) The DS-TWR protocol proposed 10 years ago was originally designed for stationary ranging.

(b) If agents move, it suffers from ranging latency problem, like motion blur in photography.

Figure 2: Enabling dynamic swarm ranging requires a fundamental theoretical breakthrough in a decade-old, well-established, and widely adopted ranging theory, making it both technically challenging and highly consequential.

aerial robots, which typically operate in high dynamics and collaborate to accomplish complex tasks in groups, such as search and rescue, cooperative navigation, and multi-robot manipulation. Consequently, accurate and reliable ranging becomes critical, as it serves as the foundation for localization, formation maintenance, and collision avoidance.

However, the classical DS-TWR protocol was originally designed for stationary one-to-one ranging. How can it be extended to many-to-many swarm ranging, and how can it support highly dynamic agents? This paper focuses on the second question, as the first one has been largely addressed in recent work [20, 23, 36, 37]. Their main idea is to periodically broadcast ranging messages, so as to exchange timestamps efficiently through the shared wireless channel. We refer to this method as the *periodic DS-TWR*, which computes the distance with exchanged timestamps by the same theory as the classic DS-TWR. We therefore use the terms *classic DS-TWR* and *periodic DS-TWR* interchangeably. Notably, the more embodied agents participate in swarm ranging, the longer the broadcast period required to avoid channel congestion.

The periodic DS-TWR does not adequately support highly dynamic agents, as it suffers from a ranging latency problem. As illustrated in Fig. 2(a), DS-TWR requires two rounds of timestamp exchanges for each pair of agents to complete a ranging procedure. Since the timestamp exchanges via broadcasting take time, for stationary or slowly moving agents, the computed distance remains accurate. However, if agents move significantly, as shown in Fig. 2(b), the computed distance estimates may be outdated because the exchanged timestamps are delayed. This effect is analogous to motion blur in photography caused by delayed exposure. Consequently, agents may collide if such delayed distance estimates are directly used for collision avoidance.

The root cause lies in the distance computation theory underlying DS-TWR, which implicitly assumes that both agents remain stationary during the timestamp exchange process. When applied to a group of dynamically moving embodied agents, this theoretical assumption no longer holds, necessitating a new theoretical breakthrough. However, **achieving a theoretical breakthrough in a decade-old, well-established, and widely adopted ranging theory is both technically challenging and highly consequential**.

Since its introduction a decade ago, DS-TWR has coexisted with another mainstream UWB ranging technique, *TDoA*. **A long-standing question in the field is whether TWR and TDoA techniques can be integrated seamlessly to combine their strengths and mitigate their weaknesses.** TDoA estimates distance differences by measuring the differences in signal arrival times from multiple transmitters [13, 45], and supports an unlimited number of agents [10, 33]. TDoA and TWR together account for around 90% of existing ranging implementations [3]; however, they use different message formats, which makes the two methods incompatible and prevents dynamic switching between them during operation.

Main contributions are summarized as follows.

- We establish a novel *iterative convergent theory* for dynamic multi-agent ranging scenario, based on which, a brand new DSR-IC algorithm is proposed for obtaining the ToF. The DSR-IC algorithm is significantly different from DS-TWR, representing a fundamental theoretical breakthrough in a decade-old, well-established, and widely adopted ranging theory.
- We propose the DSR-CP algorithm to further compensate for delay through prediction. Based on DSR-IC and DSR-CP, we design a novel ranging protocol: **Dynamic Swarm Ranging (DSR)** protocol that directly addresses the ranging latency problem.
- We design a dedicated ranging message, ranging table (data structure) and a novel state machine and its transition that maintains ranging table updates. Moreover, our implemented ranging message format maintains compatibility with TDoA, enabling DSR and TDoA to seamlessly switch or coexist.
- Extensive experiments demonstrate that the proposed method achieves significant improvements over DS-TWR. In dynamic scenarios, DSR reduces Mean Absolute Error (Mean AE) by 37.62% and Root Mean Square Error (RMSE) by 32.10%, confirming its effectiveness and accuracy.

2 Preliminary

Ultra-wideband (UWB) is a wireless communication technology that transmits data over an extremely broad frequency spectrum with very short pulses, typically on the order of

1 ns. Such an impulse-based signaling scheme, characterized by its exceptionally brief pulse duration and distinct peak energy, enables highly accurate timestamping of signal transmission and reception, thereby allowing precise distance estimation based on ToF measurements. Commercial UWB transceiver chips generally operate in the 3.5-8 GHz band and can achieve ranging accuracy better than 10 cm by DS-TWR protocol.

2.1 DS-TWR Ranging Theory

DS-TWR, proposed in 2016 [30], standardized in the IEEE 802.15.4z-2020 [1] and FiRa white paper [2], is a widely used protocol for Time-of-Flight-based distance estimation. As illustrated in Fig. 3, a complete ranging cycle involves four message types exchanged between agents A and B in a fixed sequence: *poll*, *response*, *final*, and *report*. Since the transmission timestamp becomes available only after the corresponding packet has been sent, a dedicated *report* message is used to carry three timestamps for ToF computation. Let T_i and R_i denote the transmission and reception timestamps, respectively. Ranging computation is triggered upon receiving the *report* message, and R_e represents its reception timestamp.

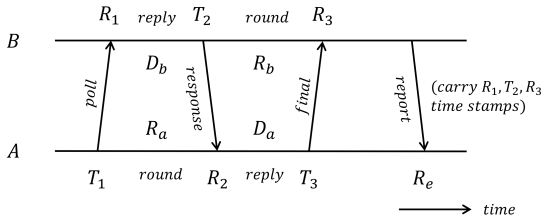


Figure 3: Double-sided two-way ranging (DS-TWR) protocol. Two agents exchange *poll*, *response*, *final*, and *report* messages sequentially to estimate ToF.

The interval from reception to the next transmission is defined as the *reply* time, while that from transmission to the next reception is the *round* time. Intervals in Fig. 3 can be expressed as:

$$R_a = R_2 - T_1, D_b = T_2 - R_1, R_b = R_3 - T_2, D_a = T_3 - R_2. \quad (1)$$

Let t denote the signal propagation time ToF. Assuming the agents remain stationary during the ranging process, the two round durations can then be written as $R_a = 2t + D_b$, $R_b = 2t + D_a$, which leads to the final DS-TWR ToF computation formula:

$$t = \frac{R_a R_b - D_a D_b}{R_a + R_b + D_a + D_b}. \quad (2)$$

Note that the derivation of Formula (2) implicitly assumes that both agents remain stationary during the ranging exchange process. Theoretically speaking, if the agents move, Formula (2) no longer holds.

Some recent works [20, 23, 36, 37] extend the classic DS-TWR to *periodic DS-TWR*, in which every agent periodically broadcasts a ranging message; however, the computation Formula (2) remains the same, and thus the problem persists.

2.2 TDoA Message Format

There are multiple implementations of TDoA [8], while we focus on the implementation provided by Bitcraze [8]. In this implementation, the tag measures the reception time differences of signals from two known anchors.

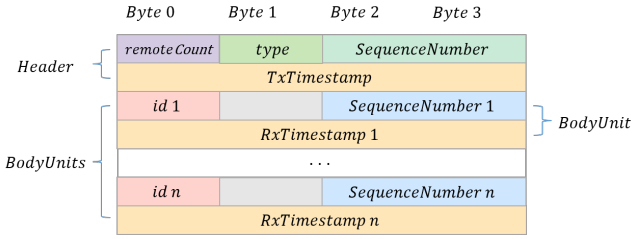


Figure 4: TDoA message format from [8].

The TDoA message format is shown in Fig. 4. Each message consists of a structured *Header* followed by a variable-length section of *BodyUnits* that aggregates measurements from remote anchors. The *Header* contains a *remoteCount* field indicating how many entries in the subsequent *BodyUnits* are populated, supporting up to eight anchors in the illustrated format. It also includes a transmission timestamp *TxTimestamp* and other information. Each *BodyUnit* entry provides information associated with one neighboring anchor. The entry includes the anchor *id*, the corresponding reception timestamp (*RxTimestamp*), and other information.

3 Theorem and Algorithms

In this section, we first analyze the inherent latency in DS-TWR measurements. Building upon this, we derive the DSR-IC and DSR-CP algorithms to resolve this latency and formulate a complete DSR algorithm that enables real-time distance acquisition.

3.1 Motivation

In this subsection, we present our motivation for revisiting the decade-old, well-established, and widely adopted ranging theory, *e.g.*, Formula (2), which forms the foundation of DS-TWR. Through an experiment, the ranging latency problem of DS-TWR is verified, as illustrated in Fig. 5.

In this experiment, we adopt a periodic DS-TWR method in which all agents broadcast messages with a predefined period, thereby enabling ranging in swarm scenarios [20, 36].

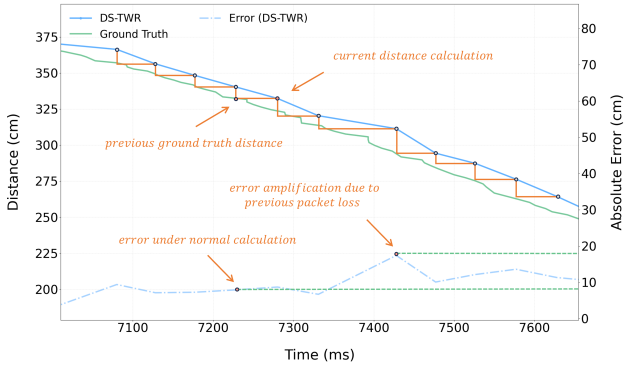


Figure 5: DS-TWR distance measurements consistently lag behind the Ground Truth in roughly one ranging interval, and packet loss amplifies the nominal ranging error as the system must rely on older timestamps.

Note that a distance calculation is triggered by the reception of a broadcast message. Fig. 6 illustrates the message exchange process, with agent B moving toward agent A.

We set the moving speed of B at approximately 1.7 m/s and the ranging period to 50 ms; thus, between two consecutive broadcasts, B moves around 8.5 cm closer. As clearly seen in Fig. 5, the DS-TWR computed distance is around 9 cm larger than the Ground Truth, which means that ranging by DS-TWR is delayed by roughly one ranging interval. Moreover, this lag is magnified when packets are lost, which is common in many-to-many swarm ranging scenarios.

One might naturally consider applying a Kalman Filter (KF) to compensate for this lag; however, this approach has notable limitations. In dynamic scenarios, the stationary assumption fails, so DS-TWR estimates do not accurately reflect the distance at t_2 [36], and any KF-based prediction is therefore grounded on a temporally misaligned reference, leading to biased estimation. Moreover, although individual KFs are computationally efficient, deploying them in dense swarms faces $O(N)$ scalability challenges, as each neighbor requires independent state and covariance management. The frequent joining and leaving of agents further exacerbates memory and management overhead, due to repeated filter re-initialization and convergence handling.

This motivates us to revisit the decade-old, well-established, and widely adopted DS-TWR ranging theory.

3.2 A Novel Iterative Convergent Theory

In this subsection, we re-derive the formula for obtaining the ToF and propose the **Double-Sided Ranging Iterative Convergent (DSR-IC)** algorithm.

As shown in Fig. 6, we consider two agents, A and B, performing double-sided ranging. The definitions of T_i, R_i, R_a, D_a, R_b, and D_b are the same as those in Fig. 3. Similarly, let

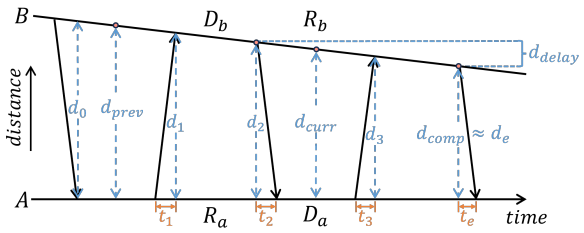


Figure 6: When B moves toward A, $t_1 \neq t_2 \neq t_3$, and thus the DS-TWR assumption no longer holds. We discover a previously unknown relation between t_{12} ($= t_1 + t_2$) and t_{23}, and derive an iterative equation between $t_{i(i+1)}$ and $t_{(i-1)i}$. The iteration is proven to converge by the novel *iterative convergent theory*.

t_i represent the ToF of the i-th message signal propagation; thus, $d_i = c \cdot t_i$ is the distance between A and B, where c denotes the speed of light. Let $t_{i(i+1)} = t_i + t_{i+1}$.

To account for clock deviation, we use the notation $\hat{(\cdot)}^{(j)}$ to denote the duration measured by agent j with clock errors, *e.g.*, $\hat{(\cdot)}^{(j)} = k_j(\cdot)$, where k_j is the clock deviation factor. We also use $\hat{(\cdot)}$ for short when no ambiguity arises. For example, $\hat{t}_{12}^{(a)} = k_a t_{12}$ and $\hat{R}_b = \hat{R}_b^{(b)} = k_b R_b$, as shown in Fig. 6.

Lemma 1. *For any two dynamically moving agents exchanging ranging messages, such as A and B in Fig. 6, we have the following relationship between $\hat{t}_{12}$ and $\hat{t}_{23}$:*

$$\hat{t}_{23}^{(a)} = \frac{\hat{R}_a \cdot \hat{R}_b - \hat{D}_a \cdot \hat{D}_b}{\hat{D}_b} - \frac{\hat{R}_b}{\hat{D}_b} \cdot \hat{t}_{12}^{(a)}, \tag{3a}$$

$$\hat{t}_{23}^{(b)} = \frac{\hat{R}_a \cdot \hat{R}_b - \hat{D}_a \cdot \hat{D}_b}{\hat{D}_a} - \frac{\hat{R}_a}{\hat{D}_a} \cdot \hat{t}_{12}^{(b)}. \tag{3b}$$

Proof. See Appendix A. □

Observation 1. *By assuming $\hat{t}_1 = \hat{t}_2 = \hat{t}_3 = \frac{\hat{t}_{12}^{(a)}}{2} = \frac{\hat{t}_{23}^{(b)}}{2}$ and adding Eqs. (3a) and (3b), the classical DS-TWR formula (2) can be directly derived. In other words, the new Eq. (3) generalizes the classical DS-TWR formula (2) to dynamic scenarios.*

We further observe that, by repeatedly applying Eq. (3), we can compute $\hat{t}_{i(i+1)}^{(a)}$ and $\hat{t}_{i(i+1)}^{(b)}$ for any i. However, since errors may be introduced in each iteration, it is important to ensure that the result converges after a finite number of iterations. Formally, we establish the following theorem.

Theorem 1 (Iterative Convergent Theory). *For any two dynamically moving agents exchanging ranging messages, such as A and B in Fig. 6, we have the following iterative relationship between $\hat{t}_{i(i+1)}$ and $\hat{t}_{(i-1)i}$:*

$$\hat{t}_{i(i+1)} = \begin{cases} \frac{\hat{R}_a \cdot \hat{R}_b - \hat{D}_a \cdot \hat{D}_b}{\hat{D}_a} - \frac{\hat{R}_a}{\hat{D}_a} \cdot \hat{t}_{(i-1)i}, & \text{if } \frac{\hat{R}_a}{\hat{D}_a} < 1, \\ \frac{\hat{R}_a \cdot \hat{R}_b - \hat{D}_a \cdot \hat{D}_b}{\hat{D}_b} - \frac{\hat{R}_b}{\hat{D}_b} \cdot \hat{t}_{(i-1)i}, & \text{if } \frac{\hat{R}_b}{\hat{D}_b} < 1. \end{cases} \tag{4}$$

This iteration is convergent, and $d_{curr} = \frac{1}{2}\hat{t}_{i(i+1)}c$ computes the distance, where c denotes the speed of light.

Proof. See Appendix B. □

Other than the above theoretical proof, the convergence of Eq. (4) in Theorem 1 is also verified experimentally, and the details can be found in Subsection 6.1.

Since the initial value of the iteration is not directly available, the traditional DS-TWR method is used to compute the initial estimate. In subsequent iterations, the previously computed $\hat{t}_{(i-1)i}$ is used to compute the next $\hat{t}_{i(i+1)}$.

Following the above derivations, the DSR-IC algorithm can be implemented as summarized in Algorithm 1:

Algorithm 1 DSR-IC($T_1, T_2, T_3, R_1, R_2, R_3, t_{12}$)

1: $R_a \leftarrow R_2 - T_1, D_a \leftarrow T_3 - R_2, R_b \leftarrow R_3 - T_2, D_b \leftarrow T_2 - R_1$
2: **if** $R_a/D_a < 1$ **then**
3: $\quad t_{23} \leftarrow \frac{R_a \cdot R_b - D_a \cdot D_b}{D_a} - \frac{R_a}{D_a} \cdot t_{12}$
4: **else if** $R_b/D_b < 1$ **then**
5: $\quad t_{23} \leftarrow \frac{R_a \cdot R_b - D_a \cdot D_b}{D_b} - \frac{R_b}{D_b} \cdot t_{12}$
6: **else**
7: $\quad t_{23} \leftarrow 2 \cdot \frac{R_a \cdot R_b - D_a \cdot D_b}{R_a + R_b + D_a + D_b}$
8: **end if**
9: **return** $\frac{c}{2} \cdot t_{23}$

Since the algorithm works in iteration, after the last calculation, t_{01} is available rather than t_{12}. Therefore, t_{12} is first computed from t_{01} when the computation is triggered by message reception, and t_{23} is then computed to obtain the distance estimate, as shown in Algorithm 3.

Clearly, the DSR-IC algorithm based on the *Iterative Convergent Theory* is significantly different from the widely adopted DS-TWR algorithm. The proposed *Iterative Convergent Theory* improves the ranging results for dynamic scenarios theoretically and represents a major breakthrough in a decade.

3.3 Further Compensation and Prediction

In this subsection, we propose the **Double-Sided Ranging with Compensation and Prediction (DSR-CP)** algorithm, which incorporates historical distance measurements d_{prev} to predict the lag-induced error of the computed distance d_{curr} relative to the true distance and compensate for this distance discrepancy.

The distance computed in Line 9 of Algorithm 1 is actually

$$d_{curr} = \frac{\hat{d}_2 + \hat{d}_3}{2}, \tag{5}$$

as illustrated in Fig. 6. Compared with d_e, d_{curr} still exhibits a gap. Meanwhile, the previously computed distance d_{prev} can be expressed as

$$d_{prev} = \frac{\hat{d}_0 + \hat{d}_1}{2}. \tag{6}$$

To make a further compensation and prediction, we need an assumption on the motion pattern.

Assumption 1 (Uniform Linear Motion). *We assume that each agent moves approximately in uniform linear motion during one ranging interval, which typically lasts a few dozen milliseconds.*

Now, we use d_{prev} to compensate for the gap towards d_e measured at time R_e, which is denoted as d_{comp}.

Lemma 2 (Compensation). *For any two dynamically moving agents exchanging ranging messages, such as A and B in Fig. 6, we estimate the distance d_e at time R_e by the corrected compensation distance d_{comp} as follows:*

$$d_{comp} = d_{curr} + \frac{(2R_e - R_2 - T_3) \cdot (d_{curr} - d_{prev})}{R_e - R_0}. \tag{7}$$

Proof. See Appendix C. □

Each time a message is received, *e.g.,* at time R_e, the compensation distance d_{comp} is estimated by Eq. (7). This approach has a limitation: distance can only be computed once in a ranging interval, upon a message reception. When distance information is required by the upper layer application at an arbitrary time, only the previously computed value can be returned, which results in further delay.

We therefore extend the computation of *compensation distance* at time R_e to *prediction distance* at an arbitrary time $R_t \in [R_e, R_{e+1})$. Let d_{pred} denote this distance estimation.

Lemma 3 (Prediction). *For any two dynamically moving agents exchanging ranging messages, such as A and B in Fig. 6, we estimate the distance d_t at time $R_t \in [R_e, R_{e+1})$ by the predicted distance d_{pred} as follows:*

$$d_{pred} = d_{curr} + \frac{(2R_t - R_2 - T_3) \cdot (d_{curr} - d_{prev})}{R_e - R_0}. \tag{8}$$

Proof. See Appendix D. □

However, the compensation and prediction rely on Assumption 1, while it does not quite hold under high-acceleration, low-frequency sampling conditions. To ensure the accuracy of the estimation, compensation and prediction should be deactivated under the following conditions:

- **Sharp Turns** (*status.turn*): Unsuitability of linear compensation for high acceleration.
- **Consecutive Data Loss** (*status.loss*): Decreased reliability of historical measurements.
- **Static States** (*status.static*): Risk of noise injection into inherently stable estimates.

Considering these operational constraints along with the derivation, DSR-CP can be implemented as summarized in Algorithm 2.

Algorithm 2 DSR-CP($T_3, R_0, R_2, R_e, R_t, d_{curr}, d_{prev}, status$)

if not *status.loss* **and** not *.static* **and** not *.turn* **then**
  return $d_{curr} + \frac{(2R_t - R_2 - T_3)\cdot(d_{curr} - d_{prev})}{R_e - R_0}$
end if
return d_{curr}

So far, we have introduced the two core components of DSR algorithm: DSR-IC and DSR-CP. As discussed in Subsection 3.2, the unavailability of t_{12} upon message reception forces the system to iteratively execute DSR-IC twice to obtain d_{curr}, and subsequently apply DSR-CP to output the final predicted distance. Algorithm 3 details the complete implementation of this procedure.

Algorithm 3 DSR($T_{0\ldots3}, R_{0\ldots3}, R_e, R_t, t_{01}, status$)

$d_{temp} \leftarrow$ DSR-IC($T_0, T_1, T_2, R_0, R_1, R_2, t_{01}$)
$d_{curr} \leftarrow$ DSR-IC($T_1, T_2, T_3, R_1, R_2, R_3, 2\frac{d_{temp}}{c}$)
return DSR-CP($T_3, R_0, R_2, R_e, R_t, d_{curr}, \frac{c}{2}t_{01}, status$)

4 DSR Protocol

The previous section introduces the theorem and algorithm, and this section describes a detailed DSR protocol based on such theorem and algorithm.

Algorithm 3 computes the distance between two agents, while requiring timestamps from both sides, *e.g.*, T_0, T_1, T_2, T_3, R_0, R_1, R_2, R_3, R_e, and R_t, as well as the previously computed ToF t_{01}. Therefore, **ranging messages** must be carefully designed to carry and exchange timestamps; the **ranging table** (data structure) must be carefully designed to store timestamps and ToFs; moreover, the **state machine**, which describes the process of updating the ranging table, must be specifically designed.

Targeting ranging amongst a group of agents, the DSR protocol is designed with many-to-many swarm ranging capability in mind, as well as with consideration of **TDoA compatibility**.

4.1 Ranging Message Design

One of the most important tasks of the ranging message is to exchange local timestamps with others so that distance can be computed. To support many-to-many swarm ranging, an agent periodically broadcasts a ranging message, which carries the receive (RX) and transmit (TX) timestamps.

More specifically, the proposed structure of the ranging message is shown in Fig. 7, in which the *Header* and the *BodyUnits* are the two main parts.

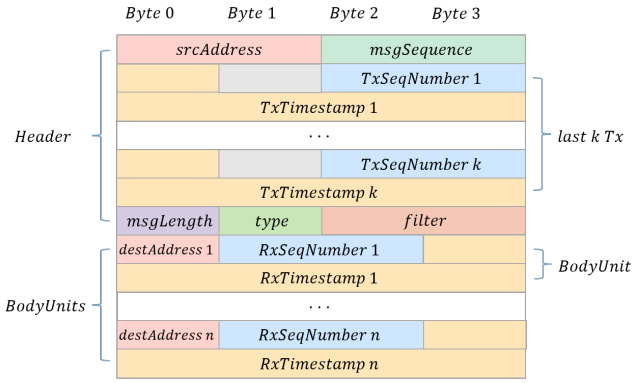


Figure 7: Ranging Message Format.

The *Header* contains the timestamps of the most recent messages transmitted by the current agent (*TxTimestamps*). The *TxTimestamps* of the current ranging message is not included, because it is not ready until this message has been transmitted. As a result, this timestamp is instead carried in the next outgoing message. Meanwhile, a boolean filter field (*filter*) is introduced to reduce processing workload, especially in a swarm ranging scenario. Whenever a message is received, the receiver combines the *filter* with its local address through a bitwise operation to quickly determine whether the message's *BodyUnits* contains timestamps relevant to itself. Only if the *BodyUnits* contains information of interest is it processed, thereby reducing the workload.

The *BodyUnits* contains a set of *BodyUnit*s, one for each neighboring agent, so as to support many-to-many swarm ranging. Each *BodyUnit* contains the local reception timestamp of the last message from one neighbor (*RxTimestamps*), along with other relevant information. When there are so many neighbors that the *BodyUnits* does not have enough room due to the ranging message payload size limitation, the question of which neighbors' information should be carried in each message arises. To address this issue, we adopt a circular-queue-based priority system to manage neighbor selection, where neighbors at the head of the queue have the highest priority. This circular queue is updated by the following rules:

- **Queue insertion:** Newly discovered neighbors are appended to the tail of the circular queue.
- **Priority demotion:** Once a neighbor's information is carried in an outgoing message, its priority is demoted to the lowest level by moving it to the tail of the queue.

- **Silence-based removal:** Each agent periodically cleans its neighbor set, removing neighbors that have not sent any packet for several periods.

During message generation, neighbors to be included in the packet are selected according to the above rules.

4.2 Ranging Table Design

Because of its iterative nature, our proposed DSR Algorithm computes the ToF continuously, upon any ranging message arrival. And in computation, it needs new timestamps, previous timestamps, and the ToF. Hence, a data structure named *ranging table* is proposed to handle this rolling information.

t_{01}^p		t_{01}				
T_0^p	R_1^p	T_0	R_1	T_2	R_3	$state$
R_0^p	T_1^p	R_0	T_1	R_2	T_3	R_e

Figure 8: Ranging Table Data Structure.

One ranging table is created for each neighbor agent, and all the timestamps of messages to and from that neighbor are stored here, as illustrated in Fig. 8. T_i and R_i denote the transmission and reception timestamps of messages, while t_{01} represents the ToF resulting from the previous invoke of the DSR algorithm. The backup fields (T_i^p, R_i^p, t_{01}^p) store previous timestamps and ToF, allowing the computation to fall back to valid data in case of timestamp disorder.

The update of the ranging table for any neighbor is triggered by either message transmission to or message reception from that neighbor. The following subsection provides more details.

4.3 State Machine and Its Transition Design

We design a state machine to handle the transmission and reception of messages as well as ranging table updates in the DSR protocol.

There are several possible states of the ranging table, depending on whether its cells contain information or remain empty. We summarize nine main states (S0–S8) and a few temporary states (red T). All these states are grouped into three phases, as illustrated in Fig. 9. For clarity, the backup fields in the ranging table are omitted from the figure, as they are rarely used. These fields are only accessed after the main loop begins, when message reception causes the table to become out-of-order.

There are three events that drive the transition of these states:

(1) Transmitting a ranging message;
(2) Receiving a ranging message whose *BodyUnits* carries *RxTimestamps* of a message from the current agent;
(3) Receiving a ranging message whose *BodyUnits* does not carry *RxTimestamps* of any message from the current agent.

Once any of the events occurs, the RX and TX timestamps of the current message, as well as the *RxTimestamps* carried within the message, are obtained. These timestamps are used to update the ranging table according to Fig. 9.

Since the Main Loop Phase makes up the major part of the actual state transitions, let us examine it carefully. First, upon reaching State S5, sufficient timestamps are available to compute the initial value of t_{01} using the DS-TWR algorithm, and the system thus enters the Main Loop Phase. In this phase, when events (1) and (2) alternatively occur as expected, the distance is repeatedly computed by the DSR algorithm. However, in other exceptional cases, the DSR-IC algorithm is used to obtain $\frac{t_{12}\cdot c}{2}$, which is then directly taken as the final ranging result.

Observation 2. *The DSR protocol maximizes the number of ranges during the Main Loop Phase, since any message reception yields a distance computation, either by the DSR or DSR-IC algorithm.*

It is worth noting that when a neighbor is removed by silence-based removal, the ranging table is cleared and the state machine reverts to State S0.

4.4 TDoA Compatibility Support

TWR and TDoA have long been separate, each with its own advantages. A long-standing question in the field is whether the two techniques can be integrated seamlessly to combine their strengths and mitigate their weaknesses.

Although a pure TDoA system can theoretically support an unbounded number of tags through anchor broadcasting [26], it is difficult to deploy sufficient anchors to ensure reliable coverage for a large-scale swarm in outdoor environments [18]. On the other hand, while the DSR protocol enables flexible two-way ranging without relying on anchor deployment, the number of participating agents is constrained by the message length, which limits the number of BodyUnits that can be carried. By integrating TDoA and DSR, the system combines TDoA's scalability with DSR's deployment flexibility, thereby overcoming the limitations of both.

Both DSR and TDoA-based systems infer distances from timestamped packet exchanges, leading to semantic compatibility between DSR and TDoA. As illustrated in Fig. 4 and Fig. 7, in our DSR protocol, the *txTimestamp* field provides the transmission timestamp required by TDoA, while the *remoteCount* field in TDoA is equivalent to the number of BodyUnits in a DSR packet, a value that can be derived from the *msgLength* field. The *destAddress*, *RxSeqNumber*, and *RxTimestamp* fields within each BodyUnit correspond

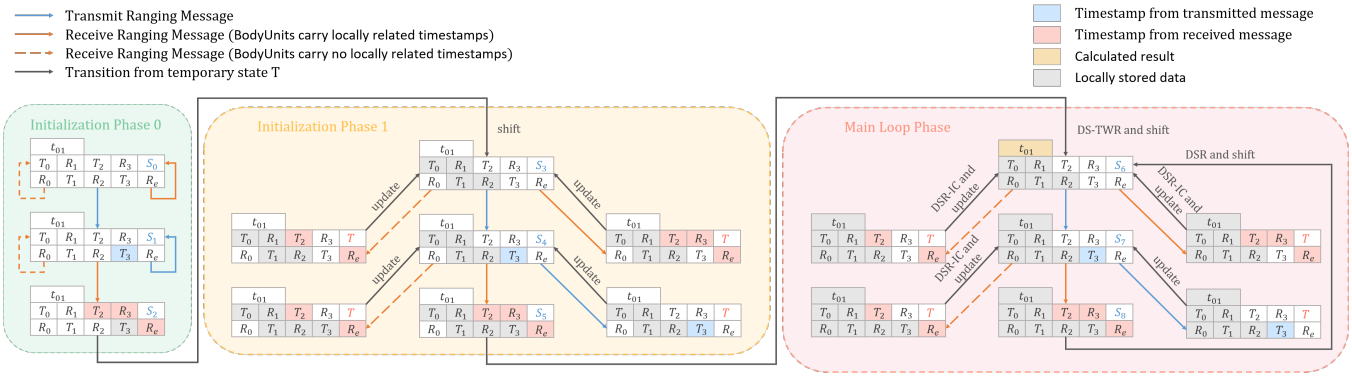


Figure 9: A state machine governs updates of the ranging table in the DSR protocol. Based on cell occupancy, the table is categorized into nine main states (S_0–S_8) and several temporary states (red T), grouped into three phases. State transitions are triggered by three events: transmitting a ranging message, receiving a message whose *BodyUnits* contains *RxTimestamps* related to the current agent, and receiving a message without such timestamps. When any event occurs, the TX/RX timestamps and carried timestamps of the current message are extracted to update the table. Upon reaching S5, sufficient timestamps are available to compute the initial t_{01} via DS-TWR, after which the system enters the main loop where ranging is repeatedly computed using DSR or, in exceptional cases, DSR-IC.

directly to the *id*, *seq*, and *rxTimestamp* entries of each remote anchor data block in TDoA.

In this work, we focus primarily on the scenario in Fig. 10.

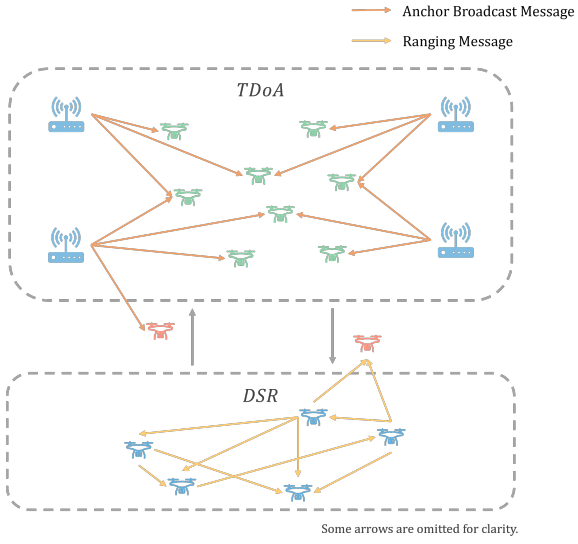


Figure 10: In a DSR-TDoA integrated system, some agents perform ranging using the DSR mode while others operate in the TDoA mode. When an agent receives a message from the other mode, the system decides based on a predefined mechanism whether it should switch modes.

To enable this integration in practice, we propose a DSR-TDoA integration mechanism that allows agents to switch seamlessly between the TDoA system and the DSR-based ranging swarm, as illustrated in Fig. 10. A *type* field is introduced in the message format to distinguish TDoA broadcast packets from DSR messages. Based on this design, the system switches between the two modes according to the following mechanism:

- In DSR mode, if an agent receives a broadcast from a TDoA anchor, it suspends DSR transmissions and ignores other DSR messages, processing only the anchor's broadcast to participate in TDoA localization.
- In TDoA mode, if an agent continuously receives DSR messages but fails to detect any anchor broadcasts for several consecutive ranging interval, it switches back to DSR mode, actively sending and receiving DSR messages to perform collaborative ranging with neighbors running the DSR protocol.

5 Implementation

We deployed the DSR protocol on the Crazyflie 2.1 micro-UAV platforms equipped with onboard DW3000 UWB transceiver chips. A single UWB channel is shared by all UAVs to support broadcast communication. To obtain accurate real-time inter-UAV distance measurements, each UAV is instrumented with a rigid-body marker plate carrying four retro-reflective markers, as shown in Fig. 11. We use a VICON system to track these rigid bodies at 100 Hz and obtain their 3D poses. The UAV positions are reconstructed from the rigid-body poses, and the pairwise inter-UAV distances are then computed.

Additionally, we deploy an auxiliary UAV equipped with a DW3000 transceiver to serve as a sniffer. This UAV continuously listens on the shared UWB channel and forwards all captured packets to a laptop via the USB cable. The captured packets are subsequently processed offline within a multi-process evaluation framework to reconstruct the in-flight

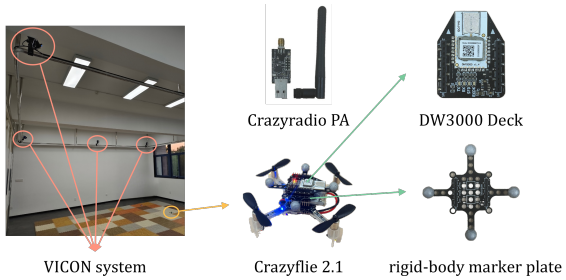


Figure 11: The protocol is deployed on Crazyflie 2.1 micro-UAV platforms, with each UAV communicating via a DW3000 UWB transceiver. Each UAV is equipped with a rigid-body marker plate, allowing the VICON system to track their 3D poses at 100 Hz and compute inter-UAV distances.

ranging process. This enables fully deterministic replay of each ranging event, ensuring that different protocols are evaluated under the same ranging procedure, thereby guaranteeing strictly fair performance comparisons.

6 Evaluation

In this section, we comprehensively evaluate the effectiveness of the proposed DSR protocol. We validate the *Iterative Convergent Theory* and assess the protocol's performance across different scenarios, using the IEEE 802.15.4z standard [1] and the recently proposed swarm ranging protocol SRv2 [20] as baselines. In addition, TDoA-compatible ranging experiments confirm the feasibility of the DSR-TDoA integration mechanism.

6.1 *Iterative Convergent Theory* Validation

The *iterative convergent theory* in Subsection 3.2 has been theoretically proved in Appendix B, and this subsection further verifies its convergence experimentally.

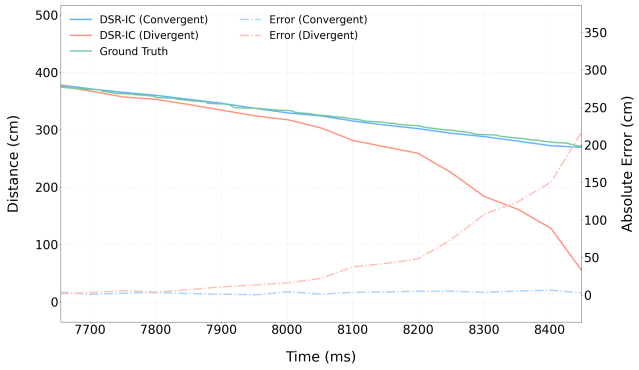


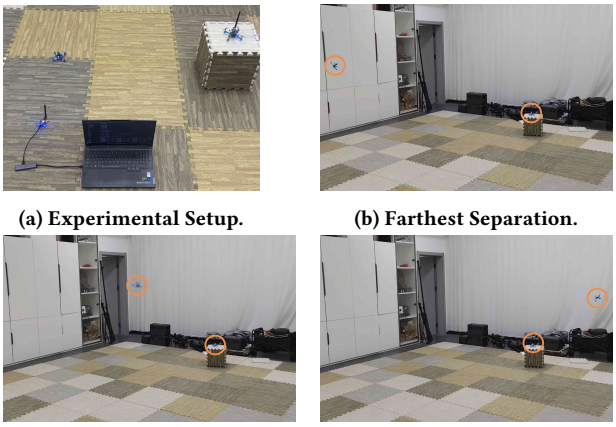
Figure 12: Failure to select the computation formula corresponding to the conditions that satisfy the specified ratio will lead to divergent accumulation of ranging errors.

Eq. (4) in the theorem has two cases, determined by the **if** condition, *e.g.*, each condition corresponds to one case. To illustrate the necessity of this matching, we conduct a comparative experiment with two treatments: (1) conditions correctly matched to the cases in Eq. (4), and (2) a condition intentionally matched to the opposite case. They are labeled DSR-IC (Convergent) and DSR-IC (Divergent), respectively. Fig. 12 presents the experimental results.

It can be observed that, over time, the curve under the divergent treatment progressively deviates from the VICON reference. In contrast, the convergent treatment curve maintains accurate performance, and the error curve remains stable. These results thus provide a clear validation of the convergence of Eq. (4) in Theorem 1.

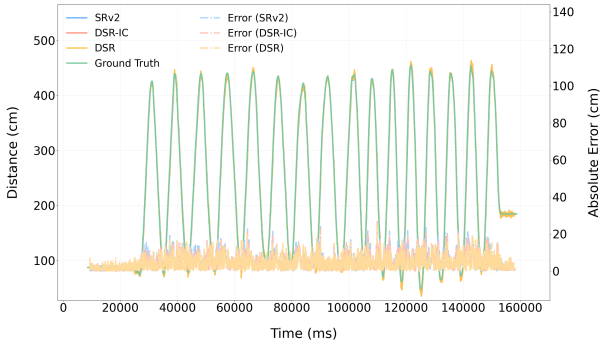
6.2 Dynamic Ranging Experiments

To evaluate the performance of DSR and demonstrate the contributions of DSR-IC and DSR-CP, we set up a two-UAV scenario (Fig. 13) and performed flight-based ranging experiments with a 50 ms measurement interval.



(a) Experimental Setup. **(b) Farthest Separation.**

(c) Mid-Flight. **(d) Closest Separation.**



(e) Distance Variation over Time.

Figure 13: A two-UAV scenario with one mobile UAV performing back-and-forth flight and one stationary reference UAV.

A total of 18,000 measurements, comprising 6,000 samples per method, were conducted to assess the ranging performance of SRv2, DSR-IC, and DSR. Detailed comparative data are provided in Fig. 14 and Table 1.

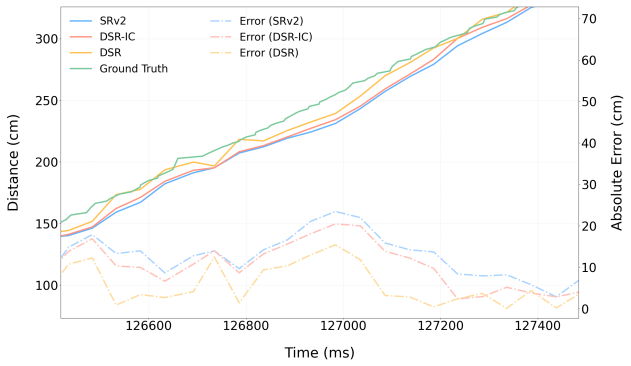


Figure 14: Distance Measurement Results and Errors.

The figure illustrates partial ranging results for a scenario where UAVs move apart at approximately 2 m/s. As shown, DSR-IC tracks the Ground Truth more accurately than SRv2, although a residual lag persists, which is consistent with the analysis in Subsection 3.3. In contrast, the DSR curve exhibits the best performance, aligning closely with the VICON-measured Ground Truth.

Table 1: Ranging Performance Comparison.

Protocol	Metrics (cm / %)				Reduction (%)			
	Mean AE	Max AE	RMSE	MRE	Mean AE	Max AE	RMSE	MRE
SRv2	6.287	26.555	7.860	3.483	-	-	-	-
DSR-IC	5.637	26.041	7.082	3.175	10.3% ↓	1.9% ↓	9.9% ↓	8.9% ↓
DSR	4.438	26.520	5.768	2.626	29.4% ↓	0.1% ↓	26.6% ↓	24.6% ↓

From the perspective of global ranging error, DSR-IC achieves lower Mean AE, RMSE, and MRE compared to SRv2, while DSR provides a 25-30% improvement in ranging accuracy. To provide a more intuitive comparison of the three ranging methods, we plot their error distributions using kernel density estimation (KDE) in Fig. 15.

As illustrated, DSR achieves the lowest 90th-percentile error (9.6 cm) and exhibits a more tightly concentrated error distribution in the low-error range compared to SRv2.

Furthermore, the ranging process is categorized into two distinct regimes based on a relative velocity threshold of 60 cm/s, encompassing different motion states:

(1) **High-speed regime**: UAVs move toward or away from each other with high relative velocity.
(2) **Low-speed regime**: UAVs exhibit low relative velocity, covering slow motion, stationary states, and the directional reversals during back-and-forth movement.

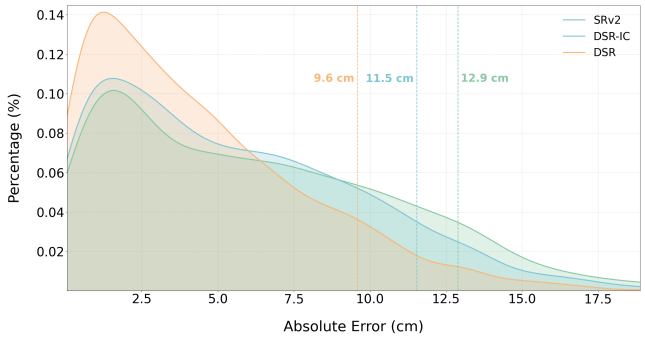


Figure 15: Kernel density estimate of ranging error for different methods with 90% quantile lines.

The corresponding ranging error statistics for both regimes are summarized in Table 2.

Table 2: Ranging Performance under Different Relative Motion Speeds.

Protocol	Metrics (cm / %)				Reduction (%)			
	Mean AE	Max AE	RMSE	MRE	Mean AE	Max AE	RMSE	MRE
Low-speed regime								
SRv2	3.525	19.028	4.716	2.798	-	-	-	-
DSR	3.467	20.028	4.696	2.801	1.67% ↓	-5.25% ↑	0.42% ↓	-0.11% ↑
High-speed regime								
SRv2	8.197	26.555	9.441	3.957	-	-	-	-
DSR	5.117	26.520	6.412	2.508	37.62% ↓	0.13% ↓	32.10% ↓	36.7% ↓

As shown, although both DSR and SRv2 exhibit noticeably larger ranging errors in the high-speed regime—indicating a significant impact of motion on accuracy—DSR consistently outperforms SRv2 under these demanding conditions. In contrast, the performance of both protocols is equivalent under low-speed conditions, with only marginal differences observed.

Overall, these results demonstrate that DSR not only effectively mitigates lag-related errors but also delivers superior accuracy in dynamic scenarios.

6.3 Real-Time Ranging Experiments

Unlike traditional ranging methods, DSR allows distance computation at an arbitrary time. As illustrated in Fig. 16, when a ranging estimate is required between two consecutive packet receptions, the system does not need to fall back to the previous measurement.

To evaluate the real-time ranging performance of DSR against SRv2, we perform distance estimations at varying sampling frequencies of 1, 3, 5, 7, and 9 per ranging interval. The corresponding results are illustrated in Fig. 17.

Since accuracy is primarily rooted in packet-reception calculations, increasing the sampling frequency elevates the ratio of estimated points, which inevitably drives up the overall

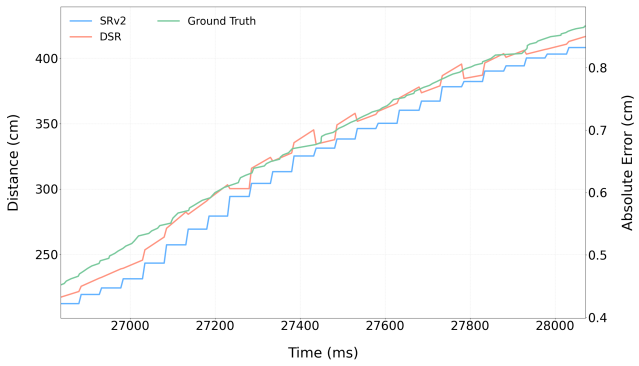


Figure 16: Real-time distance measurement results at 9 acquisitions per ranging interval, where SRv2 exhibits a staircase-like trajectory, while DSR shows a smoother curve.

ranging errors for both methods. However, DSR consistently outperforms SRv2 across all metrics. Notably, while SRv2's error degrades rapidly as calculation frequency increases from 1 to 3, DSR maintains a stable performance trajectory. Even at the highest frequency of 9 calculations per ranging interval, DSR still maintains nearly a 30% reduction in Mean AE, RMSE, and MRE compared to SRv2.

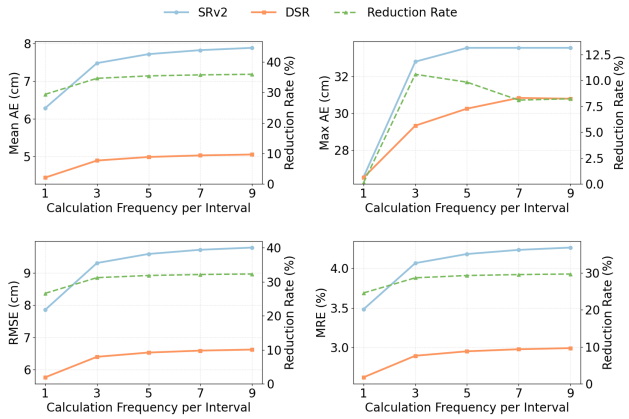


Figure 17: Real-Time Ranging Errors Line Plot under Different Sampling Frequencies.

6.4 Swarm Ranging Experiments

Subsection 4.1 introduces a circular-queue-based priority system to mitigate limitations on BodyUnits caused by constrained UWB payload in many-to-many swarm ranging. To quantify the impact of neighbor count and BodyUnit capacity on DSR, experiments are conducted by varying their ratio. For intuition, a ratio of 5 with 5 BodyUnits per rangingMessage implies 25 concurrent neighbors (*i.e.,* a 26-UAV swarm). We adopt this ratio as the independent variable, since swarm size alone—without accounting for payload constraints—cannot objectively characterize the swarm ranging performance. The IEEE 802.15.4z standard [1] and the recently proposed swarm ranging protocol SRv2 [36] serve as baselines.

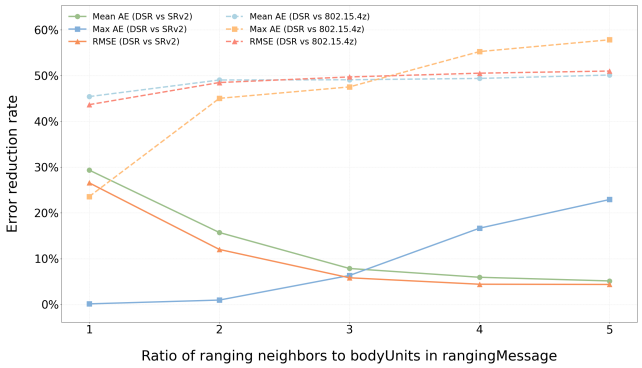


Figure 18: Error Reduction Rate of DSR Relative to SRv2 and IEEE 802.15.4z under Different Neighbor-BodyUnit Ratios.

As shown in Fig. 18, DSR consistently outperforms both baseline methods across all neighbor-BodyUnit ratios. Compared with the IEEE 802.15.4z standard, DSR demonstrates a distinct advantage, predominantly achieving over 40% error reduction across all metrics in multi-neighbor environments. In contrast, while DSR unequivocally outperforms SRv2, its performance lead in terms of Mean AE and RMSE narrows as the ratio scales up. As more neighbors compete for limited payload resources, the reduced per-neighbor update frequency leads to insufficient timestamp updates, thereby reducing DSR's advantage over SRv2.

6.5 TDoA-Compatible Ranging Experiments

The DSR-TDoA integration mechanism described in Subsection 4.4 is fully implemented on our experimental platform, as illustrated in Fig. 19. Our DSR-TDoA integrated system comprises two anchors (corresponding to the curves TDoA 1 and TDoA 2), one UAV executing the standard DSR protocol, and a second UAV implementing the proposed DSR-TDoA integration mechanism. Throughout the experiment, both the DSR-only UAV and the DSR-TDoA UAV remained powered on and operational.

Initially, while the anchors were inactive, the DSR-TDoA UAV joined the DSR group for ranging. Once the anchors were activated, the DSR-TDoA UAV switched to TDoA mode, suspending DSR-based ranging and transitioning to TDoA-based ranging. The resulting TDoA measurements from both anchors are approximately symmetric about the zero-axis ($y = 0$), which is consistent with the theoretical expectations.

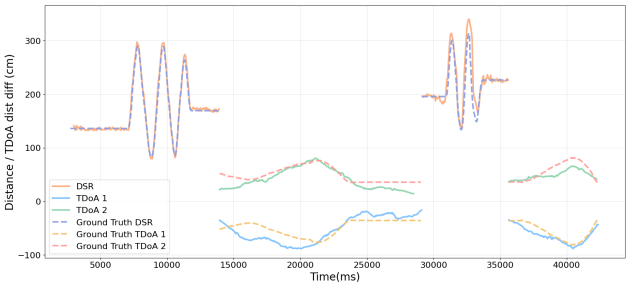


Figure 19: The UAV executing the DSR-TDoA integration mechanism seamlessly transitions between modes.

In the next phase, all anchors were sequentially deactivated, and the DSR-TDoA UAV automatically resumed DSR mode and restarted DSR-based ranging with the DSR-only UAV. Finally, when the anchors were reactivated, the DSR-TDoA UAV was able to reliably switch back to TDoA mode, demonstrating its ability to seamlessly transition between ranging modes.

7 Related Work

TWR Derivative Work: In recent years, researchers have investigated a variety of extensions and optimizations of two-way ranging (TWR), including single-sided two-way ranging (SS-TWR) [14], double-sided two-way ranging (DS-TWR) [30], symmetric DS-TWR (SDS-TWR) [24], asymmetric DS-TWR (ADS-TWR) [29, 39]. However, most existing studies on DS-TWR assume static scenarios during ranging. Even works that consider swarm settings [20, 36] do not explicitly address the impact of node mobility at the formula level. In contrast, this work extends DS-TWR-based ranging to dynamic scenarios and directly tackles motion-induced errors from a theoretical perspective.

Hybrid TDoA/DS-TWR: Several studies have attempted to combine time difference of arrival (TDoA) and DS-TWR to leverage their complementary advantages. Rathje et al. [32] and Zhu et al. [53] incorporated TDoA concepts into systems based on DS-TWR. Hong et al. [19] improved ranging accuracy by cross-validating results from TDoA and TWR, while Kolakowski et al. [22] also demonstrated that combining TWR and TDoA can enhance positioning accuracy. However, these studies typically integrate TDoA and DS-TWR in a simple manner and do not fully exploit their respective strengths. In contrast, our work unifies the message formats of both techniques, enabling seamless coexistence and flexible switching between TDoA and DS-TWR within a single system.

DS-TWR Accuracy Enhancement: Given the widespread use of DS-TWR [3], many studies have sought to reduce its ranging errors through protocol optimization and filtering techniques. Prior work has explored tuning protocol parameters to mitigate errors [34] or incorporating angle-of-arrival (AoA) measurements to improve localization accuracy [48]. Wei et al. [43] and Minh et al. [28] applied Kalman filtering in outdoor and indoor scenarios, respectively, although [28] reported only limited accuracy improvement. Similarly, Wu et al. [44] introduced Kalman filtering under non-line-of-sight (NLOS) conditions to enhance DS-TWR performance. Overall, existing efforts mainly target stochastic error reduction, while the motion-induced latency inherent to the DS-TWR formulation remains largely unaddressed.

8 Conclusion and Discussion

This paper investigates UWB-based ranging for dynamic swarms of embodied agents, tackling the long-standing latency issue of DS-TWR in highly mobile environments. We propose a novel algorithm that re-derives the ToF estimation formula and incorporates a prediction-based compensation mechanism to fundamentally mitigate this bottleneck under mobility. The proposed protocol's message format is compatible with TDoA, and we implement an integrated system with adaptive switching capabilities between DSR and TDoA. The performance of the proposed DSR protocol is validated through comparisons with the IEEE 802.15.4z standard and the recently proposed swarm ranging protocol SRv2 [20], demonstrating its superiority.

The primary advantage of DSR lies in its low-latency performance, making it particularly suitable for highly dynamic scenarios. However, like all message-based ranging protocols, DSR struggles to maintain sufficient accuracy in high-acceleration scenarios when faced with low ranging frequencies and high packet loss rates. Nevertheless, given that DSR estimates are theoretically unbiased, the protocol is robust as long as a sufficiently high ranging frequency is maintained. With the rapid advancement of hardware technology making high-frequency swarm ranging increasingly feasible, we anticipate that the performance of DSR will be further maximized in future high-rate ranging scenarios.

References

[1] 2020. IEEE Standard for Low-Rate Wireless Networks–Amendment 1: Enhanced Ultra Wideband (UWB) Physical Layers (PHYs) and Associated Ranging Techniques. *IEEE Std 802.15.4z-2020 (Amendment to IEEE Std 802.15.4-2020)* (2020), 1–174. doi:10.1109/IEEESTD.2020.9179124

[2] 2022. UWB Secure Ranging in FiRa. (2022). https://www.firaconsortium.org/resource-hub/white-papers/uwb-secure-ranging-in-fira

[3] Mohammed Faeik Ruzaij Al-Okby, Steffen Junginger, Thomas Roddelkopf, and Kerstin Thurow. 2024. UWB-based real-time indoor positioning systems: A comprehensive review. *Applied Sciences* 14, 23 (2024), 11005.

[4] Claudio Anliker, Giovanni Camurati, and Srdjan Capkun. 2023. Time for Change: How Clocks Break UWB Secure Ranging. In *32nd*

USENIX Security Symposium (USENIX Security 23). USENIX Association, Anaheim, CA, 19–36. https://www.usenix.org/conference/usenixsecurity23/presentation/anliker

[5] Apple Inc. 2026. Learn about Ultra Wideband availability. https://support.apple.com/en-hk/109512. Accessed: 2026-02-26.

[6] Rayan Armani, Changlin Qian, Jiaxi Jiang, and Christian Holz. 2024. Ultra Inertial Poser: Scalable Motion Capture and Tracking from Sparse Inertial Sensors and Ultra-Wideband Ranging. In *ACM SIGGRAPH 2024 Conference Papers* (Denver, CO, USA) *(SIGGRAPH '24)*. Association for Computing Machinery, New York, NY, USA, Article 51, 11 pages. doi:10.1145/3641519.3657465

[7] Luca Barbieri, Mattia Brambilla, Andrea Trabattoni, Stefano Mervic, and Monica Nicoli. 2021. UWB Localization in a Smart Factory: Augmentation Methods and Experimental Assessment. *IEEE Transactions on Instrumentation and Measurement* 70 (2021), 1–18. doi:10.1109/TIM.2021.3074403

[8] Bitcraze AB. 2026. TDoA3 Protocol. https://www.bitcraze.io/documentation/repository/lps-node-firmware/master/protocols/tdoa3_protocol/. Accessed: 2026-02-26.

[9] BMW Group. 2023. The 2024 BMW X5 and X6. https://www.press.bmwgroup.com/usa/article/detail/T0408460EN_US/the-2024-bmw-x5-and-x6. https://www.press.bmwgroup.com/usa/article/detail/T0408460EN_US/the-2024-bmw-x5-and-x6 Press release, published February 7, 2023.

[10] Stefano Bottigliero, Daniele Milanesio, Maurice Saccani, and Riccardo Maggiora. 2021. A Low-Cost Indoor Real-Time Locating System Based on TDOA Estimation of UWB Pulse Sequences. *IEEE Transactions on Instrumentation and Measurement* 70 (2021), 1–11. doi:10.1109/TIM.2021.3069486

[11] Yifeng Cao, Ashutosh Dhekne, and Mostafa Ammar. 2024. Utrack3d: 3d tracking using ultra-wideband (uwb) radios. In *Proceedings of the 22nd Annual International Conference on Mobile Systems, Applications and Services*. 345–358.

[12] Mahmoud Elsanhoury, Petteri Mäkelä, Janne Koljonen, Petri Välisuo, Ahm Shamsuzzoha, Timo Mantere, Mohammed Elmusrati, and Heidi Kuusniemi. 2022. Precision positioning for smart logistics using ultra-wideband technology-based indoor navigation: A review. *Ieee Access* 10 (2022), 44413–44445.

[13] Osama Ennasr and Xiaobo Tan. 2020. Time-Difference-of-Arrival (TDOA)-Based Distributed Target Localization by A Robotic Network. *IEEE Transactions on Control of Network Systems* 7, 3 (2020), 1416–1427. doi:10.1109/TCNS.2020.2979864

[14] Adrian Figueroa, Belal Al-Qudsi, Niko Joram, and Frank Ellinger. 2016. Comparison of Two-Way Ranging with FMCW and UWB Radar Systems. In *Proceedings of the 13th Workshop on Positioning, Navigation and Communications (WPNC)*. IEEE, 1–6.

[15] Nakul Garg, Irtaza Shahid, Ramanujan K Sheshadri, Karthikeyan Sundaresan, and Nirupam Roy. 2025. Large network UWB localization: algorithms and implementation. In *Proceedings of the 22nd USENIX Symposium on Networked Systems Design and Implementation* (Philadelphia, PA, USA) *(NSDI '25)*. USENIX Association, USA, Article 64, 17 pages.

[16] General Motors LLC. 2026. Cadillac Digital Key – Vehicle Support. https://www.cadillac.com/support/vehicle/security/digital-key. Accessed: 2026-02-27.

[17] Zhengxin Guo, Dongzi Wang, Linqing Gui, Biyun Sheng, Hui Cai, Fu Xiao, and Jinsong Han. 2024. UWTracking: Passive human tracking under LOS/NLOS scenarios using IR-UWB radar. *IEEE Transactions on Mobile Computing* 23, 12 (2024), 11853–11870.

[18] Wan-Ning He, Xin-Lin Huang, Zhongwei Xu, Fei Hu, Shui Yu, and Abbas Jamalipour. 2025. Low-Cost, Deterministic Train Localization With Optimal UWB Anchor Deployment. *IEEE Internet of Things Journal* 12, 24 (2025), 55255–55268. doi:10.1109/JIOT.2025.3622579

[19] Juwon Hong and Jaehoon Paul Jeong. 2025. HILS: A UWB-Based Hybrid Indoor Localization Scheme for Multi-Drone Environments. In *2025 Fifteenth International Conference on Mobile Computing and Ubiquitous Networking (ICMU)*. 1–4. doi:10.23919/ICMU65253.2025.11219135

[20] Yunxi Hou, Feng Shan, Wangxiao Mao, Jiangpeng Liu, Ye Liu, Wenjia Wu, Runqun Xiong, and Junzhou Luo. 2026. Optimal Swarm Ranging Protocol for Dynamic and Dense Ultra-Wideband Networks. In *IEEE INFOCOM 2026 - IEEE Conference on Computer Communications*. Tokyo, Japan.

[21] Wei Jiang, Zhuojian Cao, Baigen Cai, Binghao Li, and Jian Wang. 2021. Indoor and outdoor seamless positioning method using UWB enhanced multi-sensor tightly-coupled integration. *IEEE Transactions on Vehicular Technology* 70, 10 (2021), 10633–10645.

[22] Marcin Kolakowski and Vitomir Djaja-Josko. 2016. TDOA-TWR based positioning algorithm for UWB localization system. In *2016 21st International Conference on Microwave, Radar and Wireless Communications (MIKON)*. IEEE, 1–4.

[23] Shushuai Li, Feng Shan, Jiangpeng Liu, Mario Coppola, Christophe de Wagter, and Guido C. H. E. de Croon. 2025. Onboard Ranging-Based Relative Localization and Stability for Lightweight Aerial Swarms. *IEEE Robotics and Automation Letters* 10, 10 (2025), 10066–10073. doi:10.1109/LRA.2025.3597039

[24] Sang C. Lian, Michael Adams, Thomas Hörmann, et al. 2019. Numerical and Experimental Evaluation of Error Estimation for Two-Way Ranging Methods. *Sensors* 19, 3 (2019), 616.

[25] Zhicheng Luo, Qianyi Huang, Xu Chen, Rui Wang, Fan Wu, Guihai Chen, and Qian Zhang. 2024. Spectrum Sensing Everywhere: Wide-Band Spectrum Sensing With Low-Cost UWB Nodes. *IEEE/ACM Transactions on Networking* 32, 3 (2024), 2112–2127. doi:10.1109/TNET.2023.3342977

[26] Junqi Ma, Fusang Zhang, Beihong Jin, Siheng Li, and Zhi Wang. 2025. MULoc: Towards Millimeter-Accurate Localization for Unlimited UWB Tags via Anchor Overhearing. In *IEEE INFOCOM 2025 - IEEE Conference on Computer Communications*. 1–10. doi:10.1109/INFOCOM55648.2025.11044652

[27] Junqi Ma, Fusang Zhang, Beihong Jin, Chang Su, Siheng Li, Zhi Wang, and Jiazhi Ni. 2024. Push the Limit of Highly Accurate Ranging on Commercial UWB Devices. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 8, 2, Article 62 (May 2024), 27 pages. doi:10.1145/3659602

[28] Tien Le Minh and Dung Trinh Xuan. 2021. Applying Kalman filter to UWB positioning with DS-TWR method in LOS/NLOS scenarios. In *2021 International Symposium on Electrical and Electronics Engineering (ISEE)*. 95–99. doi:10.1109/ISEE51682.2021.9418707

[29] Václav Navrátil and František Vejražka. 2019. Bias and variance of asymmetric double-sided two-way ranging. *Navigation* 66, 3 (2019), 593–602.

[30] Dries Neirynck, Eric Luk, and Michael McLaughlin. 2016. An alternative double-sided two-way ranging method. In *2016 13th Workshop on Positioning, Navigation and Communications (WPNC)*. 1–4. doi:10.1109/WPNC.2016.7822844

[31] Porsche AG. 2026. Porsche Digital Key. https://www.porsche.com/international/porsche-connect/digital-key/. Accessed: 2026-02-27.

[32] Patrick Rathje and Olaf Landsiedel. 2024. Time Difference of Arrival Extraction from Two-Way Ranging. arXiv:2204.08996 [eess.SP] https://arxiv.org/abs/2204.08996

[33] Luca Santoro, Matteo Nardello, Davide Brunelli, and Daniele Fontanelli. 2023. UWB-Based Indoor Positioning System With Infinite Scalability. *IEEE Transactions on Instrumentation and Measurement* 72 (2023), 1–11. doi:10.1109/TIM.2023.3282299

[34] Mohammed Ayman Shalaby, Charles Champagne Cossette, James Richard Forbes, and Jerome Le Ny. 2024. Reducing two-way ranging variance by signal-timing optimization. *IEEE Trans. Aerospace Electron. Systems* 60, 3 (2024), 3718–3724.

[35] Mohammed Ayman Shalaby, Charles Champagne Cossette, Jerome Le Ny, and James Richard Forbes. 2024. Multi-robot Relative Pose Estimation and IMU Preintegration Using Passive UWB Transceivers. *IEEE Transactions on Robotics* 40 (2024), 2410–2429. doi:10.1109/TRO.2024.3370027

[36] Feng Shan, Haodong Huo, Jiaxin Zeng, Zengbao Li, Weiwei Wu, and Junzhou Luo. 2022. Ultra-Wideband Swarm Ranging Protocol for Dynamic and Dense Networks. *IEEE/ACM Transactions on Networking* 30, 6 (2022), 2834–2848. doi:10.1109/TNET.2022.3186071

[37] Feng Shan, Jiaxin Zeng, Zengbao Li, Junzhou Luo, and Weiwei Wu. 2021. Ultra-Wideband Swarm Ranging. In *IEEE INFOCOM 2021 - IEEE Conference on Computer Communications*. 1–10. doi:10.1109/INFOCOM42981.2021.9488717

[38] Kai Sun, Shuai Wang, Renjie Zhao, Ruofeng Liu, Weiwei Chen, Zhimeng Yin, Wei Gong, and Shuai Wang. 2025. WULoc: Achieving Extremely Long-range High-precision Localization via Wi-Fi-UWB Connection. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 9, 1, Article 18 (March 2025), 24 pages. doi:10.1145/3712282

[39] Oskar Sundin. 2021. UWB-TWR performance comparison in a hybrid node network.

[40] Unitree Technology Co., Ltd. 2023. Unitree Technology Privacy Policy. https://marketing.unitree.com/go2En/docPrivacy.html. Effective July 14, 2023; Accessed: 2026-02-26.

[41] Shuai Wang, Yunze Zeng, Vivek Jain, and Parth Pathak. 2025. *UMusic: In-car Occupancy Sensing via High-resolution UWB Power Delay Profile.* Association for Computing Machinery, New York, NY, USA, 116–129. https://doi.org/10.1145/3715014.3722049

[42] Zhi Wang, Beihong Jin, Yuhui Chen, Fusang Zhang, Siheng Li, Junqi Ma, Tao Gong, and Yu Fu. 2025. RF-AE: Single-site Arterial Elasticity Estimation Using UWB Signals. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 9, 2, Article 53 (June 2025), 26 pages. doi:10.1145/3729460

[43] Liu Wei, Wu Jian, Yin Lei, and Shen Ziyang. 2023. Optimization of Outdoor Long-Range Localization Algorithm Based on DS-TWR. In *2023 5th International Conference on Intelligent Control, Measurement and Signal Processing (ICMSP)*. 1113–1117. doi:10.1109/ICMSP58539.2023.10170886

[44] Jian Wu, Yijing Xiong, Wenyang Li, and Wenwei Xia. 2025. Adaptive Kalman Filter-Based UWB Location Tracking with Optimized DS-TWR in Workshop Non-Line-of-Sight Environments. *Sensors* 25, 24 (2025). doi:10.3390/s25247682

[45] Bin Xu, Guodong Sun, Ran Yu, and Zheng Yang. 2013. High-Accuracy TDOA-Based Localization without Time Synchronization. *IEEE Transactions on Parallel and Distributed Systems* 24, 8 (2013), 1567–1576. doi:10.1109/TPDS.2012.248

[46] Hongji Yan, You Li, Zixuan Huang, Bingpeng Zhou, Xueli Guo, and Xuehang Sun. 2026. Hierarchical Constraint Fusion for Robust 3D Rigid Body Localization Using UWB Networks. *IEEE Transactions on Mobile Computing* (2026), 1–16. doi:10.1109/TMC.2026.3654960

[47] Jing Yang, BaiShun Dong, and Jiliang Wang. 2022. VULoc: Accurate UWB Localization for Countless Targets without Synchronization. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 6, 3, Article 148 (Sept. 2022), 25 pages. doi:10.1145/3550286

[48] Di Zhang, Hongbiao Xu, Li Zhan, Ye Li, Guangqiang Yin, and Xinzhong Wang. 2025. Accurate Joint Estimation of Position and Orientation Based on Angle of Arrival and Two-Way Ranging of Ultra-Wideband Technology. *Electronics* 14, 3 (2025). doi:10.3390/electronics14030429

[49] Zijian Zhang, Hanying Zhao, Jian Wang, and Yuan Shen. 2022. Signal-Multiplexing Ranging for Network Localization. *IEEE Transactions on Wireless Communications* 21, 3 (2022), 1694–1709. doi:10.1109/TWC.2021.3106172

[50] Hanying Zhao, Lingwei Xu, Yi Li, Feiyang Wen, Haoran Gao, Changwu Liu, Jincheng Yu, Yu Wang, and Yuan Shen. 2025. Robust and Scalable Multi-Robot Localization Using Stereo UWB Arrays. *IEEE Transactions on Robotics* 41 (2025), 5645–5662. doi:10.1109/TRO.2025.3587854

[51] Hanying Zhao, Zijian Zhang, Lingwei Xu, Yu Wang, and Yuan Shen. 2024. Enhancing Timeliness in Asynchronous Vehicle Localization: A Signal-Multiplexing Network Measuring Approach. *IEEE Transactions on Intelligent Transportation Systems* 25, 10 (2024), 13729–13743. doi:10.1109/TITS.2024.3404784

[52] Hao Zhou, Kuang Yuan, Mahanth Gowda, Lili Qiu, and Jie Xiong. 2024. Rethinking orientation estimation with smartphone-equipped ultra-wideband chips. In *Proceedings of the 30th Annual International Conference on Mobile Computing and Networking*. 1045–1059.

[53] Yanping Zhu, Lei Huang, Jing Liu, Zhongkang Cao, Jinli Chen, and Zijian Mu. 2022. A Real-Time UWB Location and Tracking System Based on TWR-TDOA Estimation and a Simplified MPGA Layout Optimization. *Mobile Information Systems* 2022 (07 2022), 1–10. doi:10.1155/2022/1194169

A Proof of Lemma 1

In this section, we provide the derivation of the relationship between $\hat{t}_{12}$ and $\hat{t}_{23}$ as proposed in Lemma 1. According to Fig. 6, the true time intervals satisfy $R_a = D_b + t_{12}$ and $R_b = D_a + t_{23}$. Therefore, the measured intervals can be expressed as

$$\hat{R}_a = k_a \cdot (D_b + t_{12}) = \frac{k_a}{k_b} \cdot \hat{D}_b + k_a \cdot t_{12},$$
$$\hat{R}_b = k_b \cdot (D_a + t_{23}) = \frac{k_b}{k_a} \cdot \hat{D}_a + k_b \cdot t_{23}.$$

By rearranging terms and applying proper scaling, the above two equations can be separately transformed into

$$(\hat{R}_a - k_a \cdot t_{12}) \cdot k_b = k_a \cdot \hat{D}_b,$$
$$(t_{23} + \frac{\hat{D}_a}{k_a}) \cdot k_b = \hat{R}_b.$$

Taking the ratio of these two equations to eliminate the clock deviation factor k_b yields

$$\frac{\hat{R}_a - k_a \cdot t_{12}}{t_{23} + \frac{\hat{D}_a}{k_a}} = \frac{k_a \cdot \hat{D}_b}{\hat{R}_b},$$

which simplifies to

$$\hat{t}_{23}^{(a)} = \frac{\hat{R}_a \cdot \hat{R}_b - \hat{D}_a \cdot \hat{D}_b}{\hat{D}_b} - \frac{\hat{R}_b}{\hat{D}_b} \cdot \hat{t}_{12}^{(a)}.$$

This expression matches (3a). Finally, by symmetry, eliminating k_a instead of k_b through a similar procedure yields (3b), thereby completing the proof.

B Proof of Theorem 1

In this section, we provide the proof of Theorem 1, demonstrating that the iterative error converges. For brevity, we define $M_i^a = \frac{\hat{R}_a \cdot \hat{R}_b - \hat{D}_a \cdot \hat{D}_b}{\hat{D}_b}$, $q_i^a = \frac{\hat{R}_b}{\hat{D}_b}$, $M_i^b = \frac{\hat{R}_a \cdot \hat{R}_b - \hat{D}_a \cdot \hat{D}_b}{\hat{D}_a}$, $q_i^b = \frac{\hat{R}_a}{\hat{D}_a}$, noting that these quantities vary across iterations. With these definitions, Eq. (4) can be rewritten in a unified form as

$$\hat{t}_{i(i+1)} = M_i^j - q_i^j \hat{t}_{(i-1)i}, \quad j \in \{a, b\}.$$

By dividing both sides by the clock deviation factor k_j, the actual ToF is obtained as

$$t_{i(i+1)} = \frac{M_i^j}{k_j} - q_i^j t_{(i-1)i}, \quad j \in \{a, b\}.$$

Accordingly, the ToF estimation error can then be expressed as

$$\begin{aligned} Error_i &= \hat{t}_{i(i+1)} - t_{i(i+1)} \\ &= \left(1 - \frac{1}{k_j}\right) M_i^j - q_i^j \cdot \left(\hat{t}_{(i-1)i} - t_{(i-1)i}\right) \\ &= \Delta M_i^j - q_i^j \cdot Error_{i-1}, \quad j \in \{a, b\}, \end{aligned}$$

where $\Delta M_i^j = \left(1 - \frac{1}{k_j}\right) M_i^j$. From this recursive relation, the general form of the error is

$$\begin{aligned} Error_i = &\sum_{m=1}^{i} \left[(-1)^{i-m} \left(\prod_{l=m+1}^{i} q_l^j \right) \Delta M_m^j \right] \\ &+ (-1)^i \left(\prod_{l=1}^{i} q_l^j \right) Error_0, \quad j \in \{a, b\}. \end{aligned} \tag{9}$$

To analyze the convergence of the above recursion, we first examine the properties of q_i^j and ΔM_i^j.

Analysis of q_i^j: According to the previous definitions, the ratios q_a and q_b can be written as

$$\begin{aligned} q_a &= \frac{\hat{R}_b}{\hat{D}_b} = \frac{R_b}{D_b} = \frac{D_a + t_{23}}{D_b}, \\ q_b &= \frac{\hat{R}_a}{\hat{D}_a} = \frac{R_a}{D_a} = \frac{D_b + t_{12}}{D_a}. \end{aligned} \tag{10}$$

Since all variables D_a, D_b, t_{12}, and t_{23} are strictly positive (> 0), it follows directly that $q_i^j > 0$ for any valid ranging exchange. Next, we analyze the upper bound of q_i^j. Suppose that $q_a \geq 1$ and $q_b \geq 1$. From Eq. (10), this implies

$$D_a + t_{23} \geq D_b, \quad D_b + t_{12} \geq D_a.$$

Combining the above inequalities yields

$$-t_{23} \leq D_a - D_b \leq t_{12}. \tag{11}$$

We now evaluate the feasibility of Eq. (11) in practical UWB (Ultra-Wideband) systems by comparing the magnitudes of the relevant variables: For typical short-range communication (up to 100 m), t_{ij} is on the order of 10^{-7} to 10^{-8} seconds (nanoseconds). In contrast, D_i is governed by the task scheduling and the ranging period, typically ranging from 10^{-3} to 10^{-1} seconds (milliseconds). Therefore, the difference $|D_a - D_b|$ is extremely unlikely to fall into the nanosecond-scale window defined by $[-t_{23}, t_{12}]$, since D_a and D_b are determined by task scheduling and do not align with nanosecond-level precision. Furthermore, in the rare event that Eq. (11) is satisfied (e.g., when $D_a = D_b$ at the millisecond-level granularity), the system falls back to the traditional DS-TWR to perform the current computation (Algorithm 1). Consequently, the condition $0 < q_i^j < 1$ holds for almost all practical UWB ranging scenarios.

Analysis of ΔM_i^j: Recall from the earlier definition that

$$\Delta M_i^j = \left(1 - \frac{1}{k_j}\right) M_i^j, \quad j \in \{a, b\}. \tag{12}$$

In typical UWB-based UAV systems, the clock deviation factor k_j can be expressed as $k_j = 1 + \epsilon_j$, where the relative clock drift $|\epsilon_j|$ is on the order of 10^{-5} to 10^{-4}. This corresponds to typical quartz or temperature-compensated crystal oscillators with a frequency stability of 5–20 ppm [4]. Substituting k_j into Eq. (12) yields

$$\Delta M_i^j = \frac{\epsilon_j}{1 + \epsilon_j} \cdot M_i^j, \quad j \in \{a, b\}.$$

For the sake of derivation, we set $j = a$ and neglect the clock deviation factor, resulting in

$$\begin{aligned} \Delta M_i^a &= \frac{\epsilon_a}{1 + \epsilon_a} \cdot \frac{R_a \cdot R_b - D_a \cdot D_b}{D_b} \\ &= \frac{\epsilon_a}{1 + \epsilon_a} \cdot \frac{D_a \cdot t_{12} + D_b \cdot t_{23} + t_{12} \cdot t_{23}}{D_b}. \end{aligned}$$

The term $\frac{D_a t_{12} + D_b t_{23} + t_{12} t_{23}}{D_b}$ is of the same order of magnitude as the ToF. Given that $\frac{\epsilon_a}{1+\epsilon_a} \ll 1$, ΔM_i^a is negligible compared to the ToF, and the same holds for ΔM_i^b (*i.e.,* $\Delta M_i^j \ll t_i$).

Convergence Analysis: Having established the properties of q_i^j and ΔM_i^j, we now evaluate the asymptotic behavior of the general error form in Eq. (9). Regarding the term $(-1)^i \left(\prod_{l=1}^{i} q_l^j \right) Error_0$, the initial error $Error_0$ is obtained via DS-TWR, as mentioned in Subsection 3.2. Since $Error_0$ is inherently small compared to t_i, and it is multiplied by a product of coefficients satisfying $0 < q_l^j < 1$, this term diminishes rapidly as i increases and can be safely neglected. Next, we consider the remaining term

$$S_i = \sum_{m=1}^{i} \left[(-1)^{i-m} \left(\prod_{l=m+1}^{i} q_l^j \right) \Delta M_m^j \right], \quad j \in \{a, b\}.$$

Since $\left| \Delta M_m^j \right| \leq \Delta M_{\max}$ holds for all m, we have

$$|S_i| \leq \Delta M_{\max} \left| \sum_{m=1}^{i} \left[(-1)^{i-m} \left(\prod_{l=m+1}^{i} q_l^j \right) \right] \right|, \quad j \in \{a, b\}.$$

Because $0 < q_l^j < 1$, the product term satisfies $0 < \prod_{l=m+1}^{i} q_l^j \leq 1$, and its magnitude increases as m increases. Therefore, according to the alternating series estimation theorem, the absolute value of the series is bounded by its last term:

$$|S_i| \leq \Delta M_{\max}.$$

Since $\Delta M_{\max} \ll t_i$, it follows that $|S_i| \ll t_i$.

Therefore, the iterative error $Error_i$ is negligible compared with the ToF term t_i, which completes the proof.

C Proof of Lemma 2

In this section, we present the derivation of d_{comp} as proposed in Lemma 2. Under steady communication conditions, the message transmission frequencies of the two agents are assumed to be identical and constant. Since t_i is negligible compared with the message intervals, consecutive round durations can be considered approximately equal. Thus,

$$\hat{R}_e - \hat{T}_3 = \hat{R}_2 - \hat{T}_1.$$

By invoking Assumption 1, the relative displacements during these two intervals are considered equal, yielding

$$\hat{d}_e - \hat{d}_3 = \hat{d}_2 - \hat{d}_1. \tag{13}$$

From Eqs. (5), (6), and (13), we can deduce the following relation:

$$d_{curr} - d_{prev} = \frac{(\hat{d}_2 - \hat{d}_1) + \hat{d}_3 - \hat{d}_0}{2} = \frac{\hat{d}_e - \hat{d}_0}{2}. \tag{14}$$

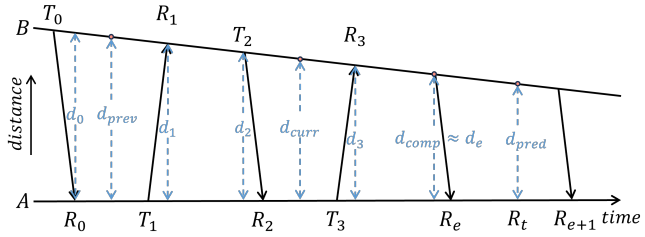


Figure 20: By leveraging the proportional relationship, we establish a correspondence between known distances and timestamps, thereby mitigating the residual gaps of DSR-IC.

Furthermore, referring to Fig. 20 and applying Assumption 1, the following proportional relationship is established:

$$\frac{d_{comp} - d_{curr}}{(R_e - R_2) - (T_3 - R_2)/2} = \frac{\hat{d}_e - \hat{d}_0}{R_e - R_0}, \tag{15}$$

where d_{comp} denotes the final estimate for the true distance d_e. By substituting Eq. (14) into Eq. (15), the estimated value of d_{comp} can be finally derived as:

$$\begin{aligned} d_{comp} &= d_{curr} + \frac{(R_e - R_2) - (T_3 - R_2)/2}{R_e - R_0} \cdot (\hat{d}_e - \hat{d}_0) \\ &= d_{curr} + \frac{(2R_e - R_2 - T_3) \cdot (d_{curr} - d_{prev})}{R_e - R_0}. \end{aligned}$$

Thus, d_{comp} serves as the compensation distance at time R_e.

D Proof of Lemma 3

In this section, we present the derivation of d_{pred} as proposed in Lemma 3. By extending the computation from time R_e to an arbitrary time $R_t \in [R_e, R_{e+1})$, we can rewrite Eq. (15) as

$$\frac{d_{pred} - d_{curr}}{(R_t - R_2) - (T_3 - R_2)/2} = \frac{\hat{d}_e - \hat{d}_0}{R_e - R_0}.$$

Similarly, the instantaneous distance at time R_t can be estimated as

$$d_{pred} = d_{curr} + \frac{(2R_t - R_2 - T_3) \cdot (d_{curr} - d_{prev})}{R_e - R_0}.$$

Hence, d_{pred} provides the prediction distance at an arbitrary time $R_t \in [R_e, R_{e+1})$.

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